

Component Scores Do Not Compose: A Held-Out Study of Clause Detection, Span Extraction and Risk Prioritization on CUAD

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Abstract—Automated contract review is usually evaluated one component at a time. We report a three-stage pipeline on the Contract Understanding Atticus Dataset (CUAD) — clause presence, span extraction, and a category-level risk layer — evaluated on a strict contract-level 80/20 split, and show that its component scores substantially overstate what the assembled system delivers. A sliding-window DistilBERT trained under multiple-instance learning loses to a full-document TF-IDF baseline (micro-F1 0.694 against 0.775). We eliminate four alternative explanations: training on all 53,662 window examples rather than a 24,000 subsample moves micro-F1 by -0.0004 ; three random seeds put the gap at 9.5 standard deviations; six aggregation rules at their own optimal thresholds all remain behind the baseline; and an untuned decision threshold accounts for 0.056 of the difference. A conjunctive ensemble reaches 0.776, but its $+0.0014$ margin is smaller than the bootstrap interval, the seed variance and the dataset-split variance alike. Run end to end, only 21.7% (95% CI [19.6, 23.9]) of gold clauses survive all three stages against the $\approx 64\%$ the component scores predict, and we localize the loss to window selection (31.7%) and a train/inference distribution mismatch. Finally, aggregate micro-F1 and the system’s stated purpose diverge: the configuration ranked first misses 37.5% of high-risk clauses while a configuration ranked last misses 13.6%. We argue that pipeline-level and risk-weighted evaluation should be reported alongside component metrics as a matter of course.

Index Terms—Legal natural language processing, contract analysis, CUAD, multiple-instance learning, model evaluation, calibration, risk assessment.

I. Introduction

COMMERCIAL contracts routinely run to tens of thousands of characters, while the provisions that determine a signer’s exposure occupy a few hundred. In CUAD [1], the median contract is 33,143 characters and the median annotated clause is 196 — roughly $16\times$ a standard transformer encoder’s input window against a target two orders of magnitude smaller. CUAD frames the resulting problem as extraction: given a category, locate the clause. It stops there, offering no indication of what a located clause means for the party signing it.

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This paper reports a system that attempts the further step and, more usefully, an evaluation of where that system fails. Our contributions are:

- 1) A negative result on long-document presence classification, with four alternative explanations tested and eliminated (Section V-B).
- 2) A demonstration that stage-wise scores do not compose: 21.7% end-to-end survival against $\approx 64\%$ predicted, decomposed into two identifiable causes (Section V-K).
- 3) Evidence that aggregate micro-F1 and user harm can invert the model ranking, measured through a risk-weighted evaluation (Section V-F).
- 4) A demonstration that a small reference set can convert a generation task into classification without the metric registering it (Section V-I).

II. Related Work

CUAD [1] supplies 510 commercial contracts with expert annotations over 41 clause categories, posed in SQuAD format [2]. Domain-adapted encoders [3] and benchmark suites [4] have advanced legal language understanding, but operationalize importance as category membership rather than as severity for a particular party. We are not aware of an established risk-severity model covering CUAD’s categories, which is why the layer we add is presented as a prototype for expert validation rather than as a validated instrument.

Long inputs are handled by retrieval, by windowing with aggregation, or by sparse attention [5]. Windowed aggregation with a maximum is the classical multiple-instance learning (MIL) rule [6]. For generation, T5 [7] and its instruction-tuned variant [8] make clause simplification natural to express; Manor and Li [9] assemble plain-English contract summaries and observe a tendency toward generic output. A substantial literature establishes that fluent abstractive output is frequently unfaithful [10], [11] and that overlap metrics such as ROUGE [12] correlate poorly with faithfulness; embedding metrics [13] improve similarity estimation without testing factuality. Our summarization result is an extreme case of the same phenomenon, and our calibration analysis follows [14].

Two properties of CUAD dominate any modelling effort on it. The first is class imbalance: per-category positive

rates span 100% to 2.5%, and the median category appears in roughly 23% of contracts. The second is length. Because contracts exceed encoder windows by more than an order of magnitude, every system must commit to a length strategy before it can commit to a model, and that choice is usually asserted rather than measured. Part of our contribution is to measure it: we report a retrieval ceiling before training and a cost benchmark for sparse attention on the target hardware, rather than arguing either point.

Evaluation practice in this area typically reports one score per component. Where pipelines are assembled, the assembled system is rarely scored end to end, so the compounding of stage errors goes unquantified — which is the gap Section V-K addresses.

III. Dataset and Protocol

All experiments use CUAD v1. We split at the contract level, 80/20, with a fixed seed: 408 training and 102 test contracts, 20,910 (contract, category) rows at a 32.1% positive rate. The same split is loaded from disk by every model, so “test” denotes the same 102 unseen contracts throughout. Positive rates per category range from 100% (Document Name) to 2.5% (Source Code Escrow); five categories have at most seven positive test contracts.

No hyperparameter was tuned. Learning rates, batch sizes and epochs are published defaults; the window geometry follows from a measurement on training data; the decision threshold is the softmax default. This protects the held-out evaluation, but means the transformer competes at conventional settings against a baseline with essentially nothing to tune — a limitation we return to in Section VII.

IV. Method

Presence. A TF-IDF baseline over the entire document feeds per-category logistic regression. In parallel, 2,000-character windows at a 1,500-character stride tile each contract; a DistilBERT [15] scores (category question, window) pairs, and the document-level score is the maximum over windows — the MIL rule. The 500-character overlap exceeds the median clause length, so no clause falls between windows. Parameter-free AND, OR and AVG rules combine the two models; nothing in them can be fitted to the test set.

Before training we measured the ceiling that a retrieval design would impose: the best query strategy places the gold clause in its selected windows only 64.0% of the time, below what the baseline already achieves. Retrieval was therefore rejected on a measurement rather than an assumption. We note the scope: this bounds lexical retrieval only.

Span. A DistilBERT question-answering head predicts start and end positions. At training time each example is re-centred so the answer begins ≈ 150 characters into a 1,200-character window; this uses a label to build a training example, never an inference input.

Risk and explanation. Each category carries a fixed High/Medium/Low level scored from the weaker party’s

TABLE I
Presence models, held-out micro-F1

Model	micro-F1	Prec.	Rec.
AND ensemble	0.776	0.772	0.780
TF-IDF baseline	0.775	0.742	0.810
Ensemble AVG	0.718	—	—
Ensemble OR	0.696	—	—
Transformer (max-pool)	0.694	0.562	0.908

TABLE II
Alternative explanations for the transformer’s deficit

Control	Effect on micro-F1	Verdict
All 53,662 windows	−0.0004 [−0.010, +0.010]	none
Random seed, 3 runs	SD 0.0085; gap 9.5 SD	none
Aggregation, 6 rules	best 0.750 < 0.775	none
Threshold, swept	+0.056 at $t=0.90$	partial

perspective (8/22/11) using five criteria: financial exposure, irreversibility, restriction of future freedom, asymmetry, and survival. This is category-level prioritization, not clause-level assessment. A FLAN-T5-small is fine-tuned to map a clause to a plain-English sentence, seeded with 41 author-written category templates.

V. Results

A. Presence Classification

Table I reports micro-F1 over the 4,182 test decisions. The windowed transformer loses to the lexical baseline by 0.081. A cluster bootstrap over contracts (10,000 resamples; the 41 decisions within a contract are not independent) puts that gap at [+0.063, +0.099], significant in every resample. The AND ensemble finishes nominally first at 0.776, with a margin of +0.0014 and an interval of [−0.0068, +0.0089] — a tie.

B. Four Controls

A bootstrap answers whether a gap could be sampling noise; it cannot answer whether the model was handicapped. Table II reports four controls.

Doubling the training data changed micro-F1 by −0.0004 while raising recall (0.908 to 0.927) and lowering precision (0.562 to 0.555): more data made the model fire more readily, not more accurately, which is what a maximum over ≈ 35 windows predicts. Reseeding moves the ensemble by more than the margin claimed for it (SD 0.0022, range 0.0042 against +0.0014). Refitting the baseline on 20 independent splits gives SD 0.0063 — the largest of the three uncertainties, and $4.5\times$ the ensemble margin.

An ablation over aggregation rules on identical window scores found that top-3 mean outperforms the maximum at the default threshold (0.7144 vs 0.6943), while the maximum is best once each rule receives its own threshold (0.7504). No rule at any threshold reaches the baseline. The threshold control is the one that qualifies our finding:

TABLE III
Aggregation ablation on identical window scores

Aggregation	F1@0.5	Best F1	at t
Max (used here)	0.6943	0.7504	0.90
Top-3 mean	0.7144	0.7210	0.40
Top-5 mean	0.6661	0.7192	0.30
Mean over all windows	0.1664	0.7114	0.10
Noisy-OR	0.5897	0.7022	0.95
Second-highest window	0.6832	0.7016	0.30
TF-IDF baseline	0.7747	—	—

TABLE IV
Uncertainty estimates against the +0.0014 ensemble margin

Source	Magnitude	$\times$ margin
Ensemble margin over baseline	0.0014	1.0
Model-seed SD (3 runs, ensemble)	0.0022	1.6
Dataset-split SD (20 splits)	0.0063	4.5
Bootstrap half-width (contracts)	0.0078	5.6

the un-chosen 0.5 boundary costs the transformer 0.056, so part of its deficit is an operating-point artefact. We do not adopt the swept threshold, since selecting it on test data would invalidate the held-out evaluation.

C. Aggregation Ablation

Max-pooling was adopted because it implements the MIL rule, not because it was compared against alternatives. Table III scores six aggregation functions on identical window scores, so only the rule varies.

Three observations follow. At the threshold actually used, the top-3 mean outperforms the maximum, recovering a fifth of the gap for a one-line change; averaging the three highest-scoring windows still asks whether evidence is concentrated somewhere, but requires it in more than one place. Given its own threshold, the maximum is the best of the six, so max-pooling is not a poor rule but one whose output distribution is mismatched to a 0.5 boundary. Most importantly, no rule at any threshold reaches the baseline, which closes the last available explanation for the result in Table I.

D. Three Sources of Uncertainty

Table IV collects the variance estimates. Dataset-split variance is the largest and the one most often omitted: model-seed variance holds the test set fixed and therefore cannot observe it. Refitting the baseline on 20 independent contract-level splits gives a mean of 0.7829 and a range of 0.0216; the split used throughout this work yields 0.7747, below that mean, so it does not flatter the baseline.

E. Rare Categories

Requiring two models to agree is a stricter bar exactly where each has least training signal. Splitting the 41 categories by training frequency, on the ten rarest (70

TABLE V
Presence performance by risk band (threshold 0.5)

Band	Model	Prec.	Rec.	Missed
High (176 pos.)	Transformer	0.404	0.841	28
	TF-IDF	0.540	0.648	62
	AND ensemble	0.585	0.625	66
	OR ensemble	0.391	0.864	24
Medium (498 pos.)	AND ensemble	0.647	0.673	163
	OR ensemble	0.423	0.922	39
Low (674 pos.)	AND ensemble	0.924	0.901	67
	OR ensemble	0.803	0.969	21

positive test decisions) the conjunctive rule scores F1 0.343 — identical to the transformer alone — by purchasing 32.5 points of precision with 25.7 points of recall, reducing detections from 35 to 17. On the remaining 31 categories the same rule gains 0.077 F1. The ensemble’s entire aggregate advantage is generated on common categories, and on five of the ten rarest it scores exactly 0.000, three of which the transformer alone was partially detecting.

F. Risk-Weighted Evaluation

Micro-F1 weights all 41 categories equally; the system’s purpose does not. Table V scores each model within each risk band.

The configuration ranked first on aggregate micro-F1 misses 37.5% of high-risk clauses; the configuration ranked last misses 13.6%. Two thirds of the aggregate’s mass is low-risk categories — dates, party names, governing law — so the metric is dominated by the clauses that matter least. On the ten rarest categories the conjunctive rule yields no F1 gain over the transformer alone (0.343 in both cases), purchasing 32.5 points of precision with 25.7 points of recall. We report this as a diagnosis rather than a retuning: selecting the OR rule after observing these test-set results would be test-set selection.

G. Threshold and Calibration

Fig. 1 shows the threshold sweep and reliability diagram. The max-pooled score has an Expected Calibration Error of 0.201 and is overconfident throughout: predictions above 0.9 correspond to the clause being present 72.8% of the time, and predictions in $[0.5, 0.6)$ to 19.5%. A maximum over dozens of windows is not a probability-preserving operation, and nothing in training asked it to be. Any interface presenting this score as confidence is misleading its user; Platt scaling on a validation split would correct it without retraining.

H. Span Extraction

Given an answer-bearing window, token-F1 is 0.763 (contract-clustered 95% CI $[0.740, 0.782]$), with 82.7% of predictions overlapping the gold span at $F1 \geq 0.5$. The clustered interval is roughly twice the width obtained by resampling clauses independently, which is the practical cost of ignoring within-contract correlation.

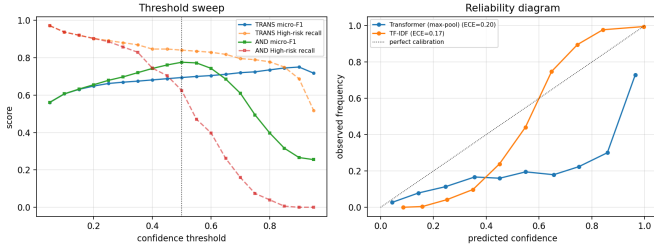


Fig. 1. Threshold sweep (left) and reliability diagram (right).

TABLE VI
ROUGE-L under two reference sets, same 150 clauses

System	References	ROUGE-L
Fine-tuned FLAN-T5	41 templates	0.775
Fine-tuned FLAN-T5	clause-specific	0.116
Zero-shot FLAN-T5	clause-specific	0.299
Template alone, no model	clause-specific	0.115

I. Summarization

CUAD contains no plain-English rewrites, so the 41 targets were author-written; every test clause is scored against one of only 41 unique strings. To separate template reproduction from summarization we scored the same 150 held-out clauses against a second, clause-specific reference set drafted independently of the model (Table VI).

The fine-tuned model emits one of the 41 templates verbatim on 100% of inputs — the correct one 70% of the time. A constant lookup table matches its score. The task became 41-way classification, and the metric did not register the change. Notably, a template-emitting model cannot hallucinate in the usual sense; it fails instead by producing fluent, confident text about a different provision, which is undetectable from the output alone.

J. Qualitative Error Analysis

Inspecting the highest-confidence mistakes identifies three recurring patterns. Lexical triggers without the obligation: a royalty clause deducting “insurance costs” is flagged as an Insurance covenant by both models (0.82/0.97) — the vocabulary is present, the obligation is not. Concept without vocabulary: a term requiring “a cancellation fee of \$5,000 per month multiplied by the number of months remaining” is textbook liquidated damages and is missed by both models (0.21/0.02), because the category’s vocabulary never appears and 24 positive training contracts are too few to learn the concept. Annotation disagreement: some rare-category “failures” are cases where the gold label is itself arguable, so the 0.000 scores on the rare tail mix model error with label ceiling effects. On the span side, failures include degenerate single-token predictions and confident answers drawn from windows that cannot contain the clause — both symptoms of an extractor with no way to abstain.

TABLE VII
End-to-end attrition over 1,348 gold clauses

Stage	Surviving	Share
Gold clauses	1,348	100.0%
Presence detected	1,052	78.0%
Span overlap ≥ 0.5	293	21.7%
Risk label correct	293	21.7%

K. End-to-End Evaluation

Running all stages in series over the 1,348 gold clauses in the test contracts gives Table VII. A clause survives only if its category is detected, the span extracted from the selected window overlaps the gold clause at token-F1 ≥ 0.5 , and the risk label is correct.

Multiplying the component figures predicts $\approx 64\%$; the pipeline delivers a third of that. Decomposing the loss over the 1,052 detected clauses: the selected window contains the gold text only 68.3% of the time, and even then extraction succeeds 39.7% of the time against 82.7% in isolation. The first cause is a detector reused as a locator — max-pooling was trained to answer whether a window contains a clause, not which window does. The second is the re-centring used at training time: legitimate with respect to leakage, but it narrows the distribution the model can read, and mean token-F1 falls from 0.763 to 0.386. Both are addressable without additional parameters.

VI. Deployment Observations

The pipeline is deployed as a local web application. Three measurements bear on whether such a system should be trusted by a non-specialist. Latency: a median-length contract takes 27.1s and the worst case 32.4s on CPU, with models resident. Truncation: a 30-window cap bounds latency but reads only the first 45,500 characters, and 38.4% of CUAD contracts exceed that — clauses beyond the cap cannot be detected, a limitation invisible in any of the offline metrics reported above. Calibration: the score presented to the user as confidence has ECE 0.201 and Brier score 0.194, so the displayed value should not be read as a probability of correctness.

Combined with the end-to-end result, this suggests a specific interface discipline: the category label and its risk badge are considerably more reliable than the quoted clause text beneath them, and an interface of this kind should present extracted spans as pointers to a location rather than as authoritative quotations.

VII. Threats to Validity

Internal. No hyperparameter was tuned, which protects the held-out evaluation but leaves the transformer untuned against a baseline with little to tune; the threshold control quantifies part of this. Construct. ROUGE-L measures its reference set rather than the task, as Section V-I demonstrates directly; micro-F1 does not measure user harm, as Section V-F demonstrates. The risk bands are our own construct and have not been

validated by anyone with legal training, so Table V inherits that limitation. External. All results come from 510 US commercial contracts drawn from SEC filings, in English. No contract outside that corpus has been evaluated. Statistical. Three variance sources were measured and all exceed the ensemble margin; three seeds is a small sample for a standard deviation; per-category statistics on the rare tail are not meaningful individually.

VIII. Discussion

The results support a general recommendation with three parts. First, a margin should be compared against its own noise before it is reported, and the relevant noise has at least three components — test-set resampling, training stochasticity, and the draw of the split itself — of which the last is both the largest here and the most frequently omitted. Second, an evaluation should be run on the assembled system, because stage-wise scores compose optimistically: ours over-predict by a factor of three, and the discrepancy localizes to interfaces between components rather than to any component. Third, the reported metric should be checked against the objective the system exists to serve; where these diverge, as in Section V-F, optimizing the metric actively degrades the system’s usefulness.

None of these checks required additional compute or data. Each required asking a question that the component-wise reporting convention does not prompt.

IX. Conclusion

On a strict held-out protocol, a full-document lexical baseline beats a windowed transformer at clause presence classification, and the result survives controls for training budget, random seed, aggregation rule and decision threshold. More consequentially, the assembled pipeline retains 21.7% of gold clauses where its component scores predict $\approx 64\%$, and the configuration that maximizes aggregate micro-F1 is the worst available choice for the clauses carrying the highest risk. We suggest that pipeline-level survival and a purpose-weighted metric belong alongside component scores in any system of this kind, and that a margin should be compared against seed and split variance — not only against a bootstrap interval — before being reported as a result.

Future work: obtaining expert validation of the risk taxonomy; a purpose-weighted selection protocol on a validation split; calibration of the confidence estimate; abstention for the span model; and faithfulness-aware evaluation of generated explanations.

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